Zane Beeai

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Education

University of Waterloo — Waterloo, ON

B.A.Sc. in Biomedical Engineering, Specialization in Computer Science, GPA: 3.92/4.0 — *Class of 2030*

Activities: AR for BME121 Digital Computation, BME181 Intro Biomedical Design, Poker Club

Technical Skills

Languages: Python, Java, JavaScript, C++, C, MATLAB, R, C#, VHDL
Software Engineering: Backend & Frontend, API Design, Version Control (Git), Concurrent & Parallel Programming
Frameworks: React, Node.js, Next.js, REST APIs, MongoDB, PyTorch, TensorFlow, NumPy, OpenCV, Simulink

Experience

Research Intern — Oct 2025 – Present
ESI Imaging — *Waterloo, ON*

- Designed and built PCB-based **antenna system**s and **acoustic sensor arrays** for low-frequency medical imaging prototypes, combining hardware design with signal processing and embedded firmware with FPGA.
- Built a rapid **antenna/metasurface design pipeline** and full **CAD–manufacturing workflow** for assembling a functional breast-imaging prototype, optimizing reflection coefficients and frequency bands.
- Implemented embedded **C++/Java firmware** for synchronized signal capture; introduced **parallelized processing** methods with FPGA that more than doubled throughput for impression-map reconstruction.

Research Assistant — Sept 2025 – Present
Advanced Concepts Research Lab (University of Waterloo) — *Waterloo, ON*

- (NDA-bound) Working on **acoustics** for **detections of biological anomalies** under Dr. Omar Rabahi.
- End-to-end development of scalable medical devices; **PCB design**, clinical testing, **signals and AI processing**.
- Designing **acoustic sensor arrays** and firmware for synchronizing **novel frequency signals**, real-time detection.

Research Assistant — July 2024 – June 2025
Sunnybrook Research Institute (FUS Lab) — *Toronto, ON*

- Researched microbubble cavitation in agarose vessels using simultaneous **optical and acoustic data** to study antivascular therapy applications under Dr. David Goertz.
- Developed a custom **MATLAB GUI** for real-time **signal processing** and image segmentation, cutting manual analysis time by **80%** and enabling high-throughput experimentation.
- Implemented **SOTA object tracking models** and novel **time-frequency transformations** to analyze micron-scale bubble dynamics, improving insight into therapeutic ultrasound mechanisms.
- Credited on thesis: X. Zhao, *Microscopy-Based Investigation of Microbubble Mediated Antivascular Ultrasound*

Research

TraceGR: Modular Spacetime Imaging using Null Geodesics in Raycasting
doi.org/10.1109/URTC65039.2024.10937597 — University of Toronto (2024)

Zane Beeai, Aditya Makkar, Dr. Ahmed Ellithy

Led research under Dr. Ahmed Ellithy to create TraceGR: a null geodesic-based, modular raycaster to produce spacetime images rivaled only by Christopher Nolan in Interstellar. Presented at **Canadian Undergraduate Math Conference (CUMC)** and **MIT-IEEE URTC**; peer-reviewed *IEEE Xplore* publication.

Projects

Voyager-O | *Three.js, WebGL, GLSL Shaders, React, Next.js, Computer Graphics, Generative AI* — /Voyager-O

- Top 40 teams/56,000 participants in NASA Space Apps Challenge. Integrated NASA/IPAC, Gaia, Star Chart datasets to render exoplanetary skies with efficient shaders, custom constellations, and generative AI stories.

WarRig X1 - Autonomous EV | *CAD, FEA, Circuit Design, Python, LiDAR, Computer Vision* — /WarRig

- Designed and built 24V electric race car from scratch. Awarded Dennis Weishar Engineering Design Award at uWaterloo's EV Challenge. Funded by Town of Oakville for $12,000 to make the car **self-driving** with CV/ML.

Exo-Ranger MK II - Exoskeleton | *C, C++, Simulink, Control Systems, Embedded, Robotics* — /exoranger-mkii

- Building 9-DOF EMG-controlled exoskeleton arm with hybrid linear/rotational actuation, real-time inverse kinematics control. Custom magnetic encoders and neuromuscular signal processing, adds 40 lb curling force.